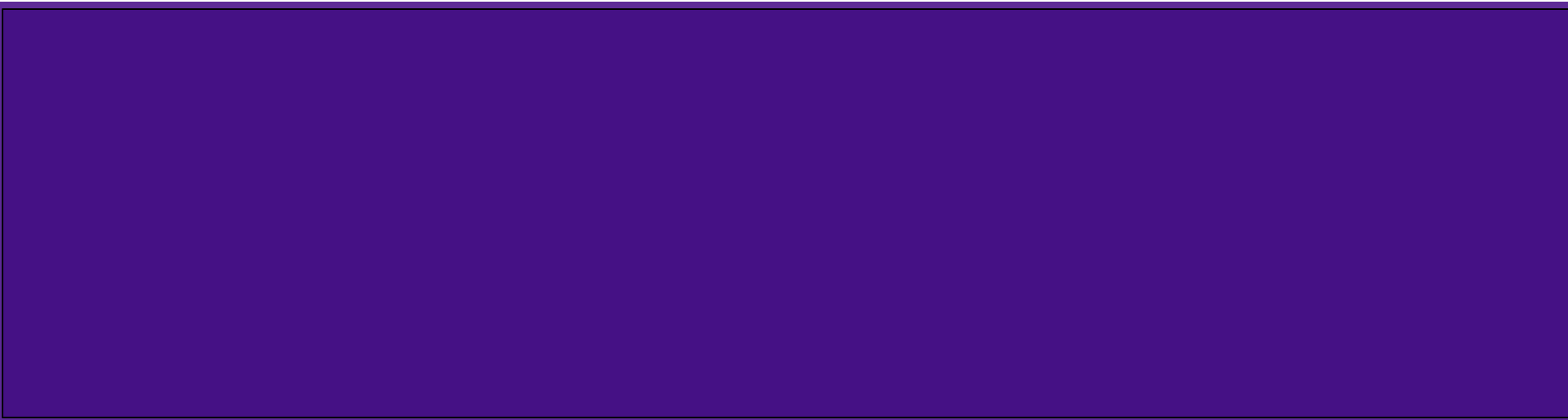




CSDE 502

Reproducibility, Replicability, & Statistical Tests

May 16, 2025



Reproducibility vs. Replicability

- **Reproducibility** is obtaining consistent results to a specific study using the same input data; computational steps, methods, and code; and conditions of analysis
- **Replicability** is obtaining consistent results across studies aimed at answering the same scientific question, each of which has obtained its own data



Reproducibility vs. Replicability

What are some components of a research study that affect the ability for someone else to reproduce or replicate it?

Reproducibility

Input data
Study design
Computational
environment

Replicability

Uncertainty
Complexity
Study design

A manifesto for reproducible science (Munafò et al. 2017)

Methods

- Protect against cognitive biases with blinding
- Require rigorous and continued training in statistics and research methods
- Increase use of independent methodological support or oversight
- Use collaborative & team science to pool resources and increase power



A manifesto for reproducible science (Munafò et al. 2017)

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A manifesto for reproducible science (Munafò et al. 2017)

Reporting and dissemination

- Increase use of study pre-registration via platforms like the Open Science Framework and Registered Reports
- Improve quality of methodological reporting with discipline-specific checklists
- Require disclosure of conflicts of interest or even exclusion of research where they exist



Replication in Social Science (Freese and Peterson, 2017)

- **Verifiability**– Do the findings presented in a study follow from the data? Aka reproducibility
- **Robustness**– Do further and different analyses of the same data provide similar results?
- **Repeatability**– Can a study's findings be observed by collecting and analyzing new data?
- **Generalizability**– Are a study's findings generalizable beyond the sample or population analyzed?



The Replication Crisis & Bad Statistics

- **P-values aren't bad, just misunderstood!!!!!!!!!!**
- Statistical tests can only answer certain types of research questions based on certain types of data.
- Statistical evidence is just that, EVIDENCE. It is not proof.

